Systems Analysis and Control

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Lecture 13: Sketching the Root Locus: Real Axis and High Gain

Overview

In this Lecture, you will learn:

Review

- Definition of Root Locus

Properties of Continuity and Symmetry

- Poles are not created or destroyed
- Root Locus is symmetric about the real axis

Points on the Real Axis

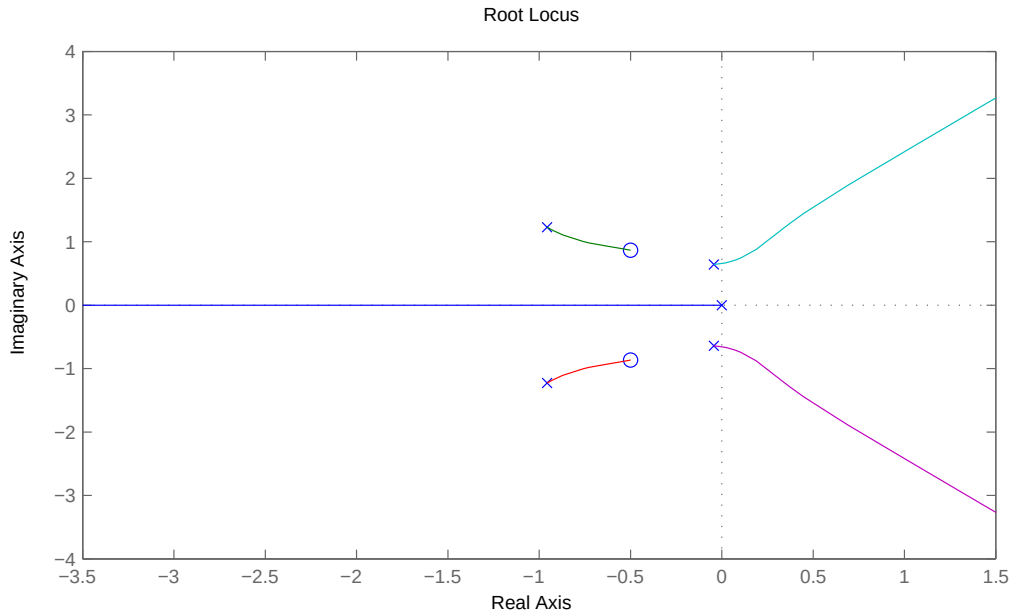
- Which parts of the real axis are on the root locus?

What Happens at High Gain?

- Where do poles go when gain increases?
- The effect of Zeros
- Asymptotes

Root Locus

Review



Definition 1.

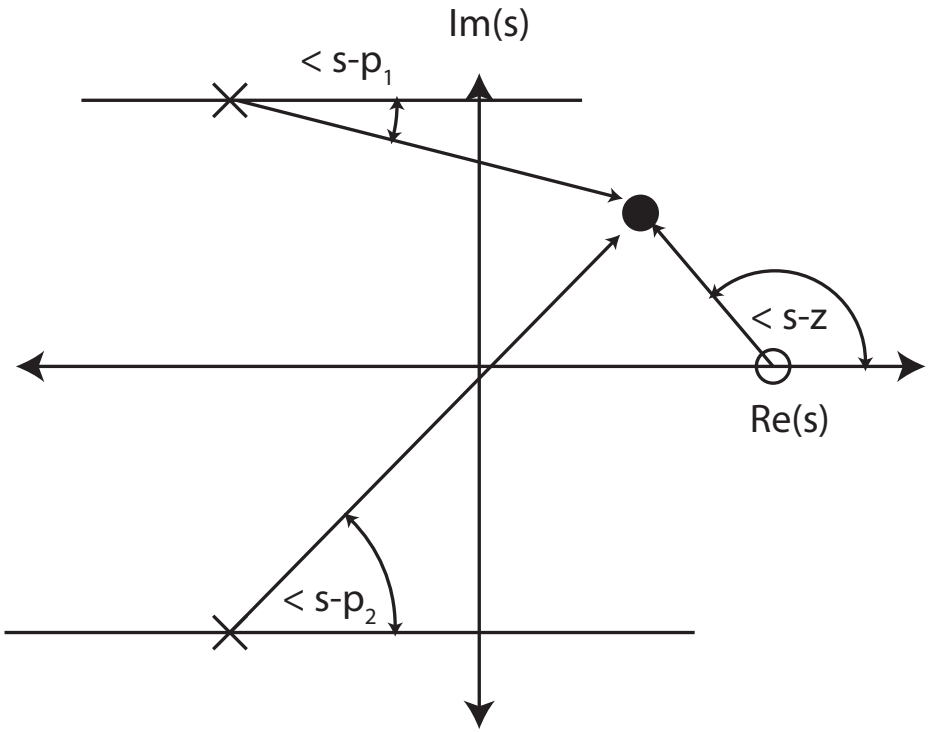
The **Root Locus** of $G(s) = \frac{n(s)}{d(s)}$ is the set of all poles of

$$\frac{kG(s)}{1 + kG(s)} = \frac{kn(s)}{d(s) + kn(s)}$$

as k ranges from 0 to ∞

Review: Definition of Root Locus

$$G(s) = \frac{(s - z_1) \cdots (s - z_m)}{(s - p_1) \cdots (s - p_n)}$$



For a point on the root locus: we need

$$\sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i) = 180^\circ \pm n \cdot 360^\circ$$

The angles from zeros **MINUS** the angles from poles have to add up to 180° (plus multiples of $\pm 360^\circ$)

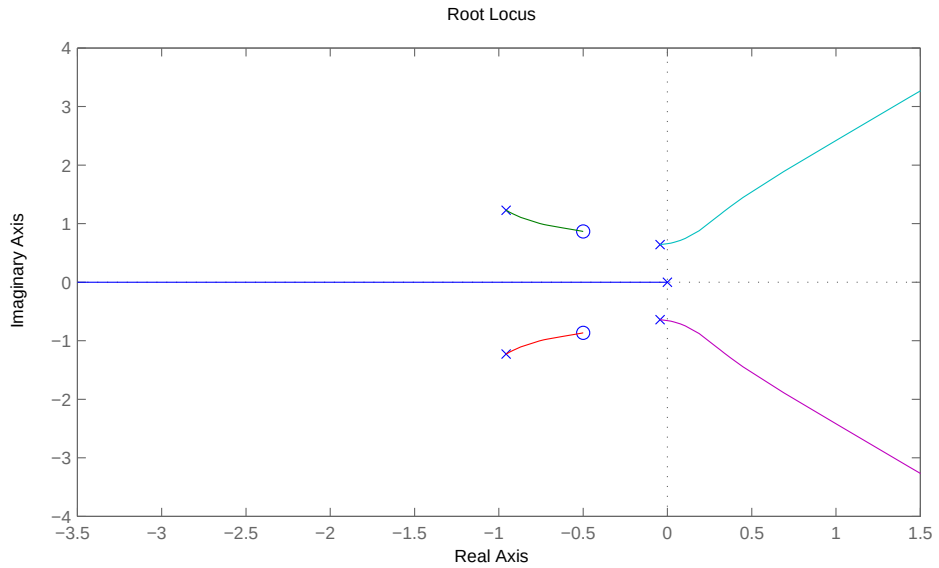
NOTE: angles are measured Counter-Clockwise (CCW)

Property of Continuity

At $k = 0$, poles start at open-loop pole locations

Continuity:

- Poles cannot be created or destroyed
- At $k = 0$, poles start at open-loop pole locations
- As k increases, the poles move around
  - But where do they go?



The Root Locus defines the motion of the poles as k increases.

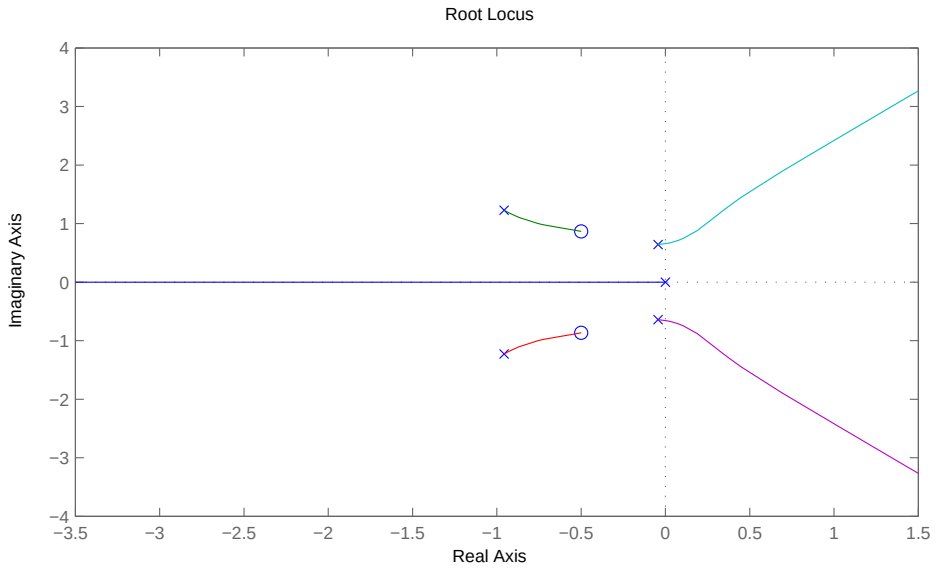
- The path of the root locus has direction.
  - Motion in the direction of increasing k.

Property of Symmetry

Constructing the Root Locus

Symmetry:

- Complex roots come in pairs: $a \pm bi$.
- Points on the root locus are mirrored above/below the real axis



We can divide points on root locus into

- Points on the real axis (easy).
- Symmetric Pairs off the real axis (harder).

Is a point on the Real Axis also on the Root Locus?

We need to determine the Phase! First find Phase Contribution from Real Zeros

What is the PHASE at a point $(s = a)$ on the Real Axis?

We need:

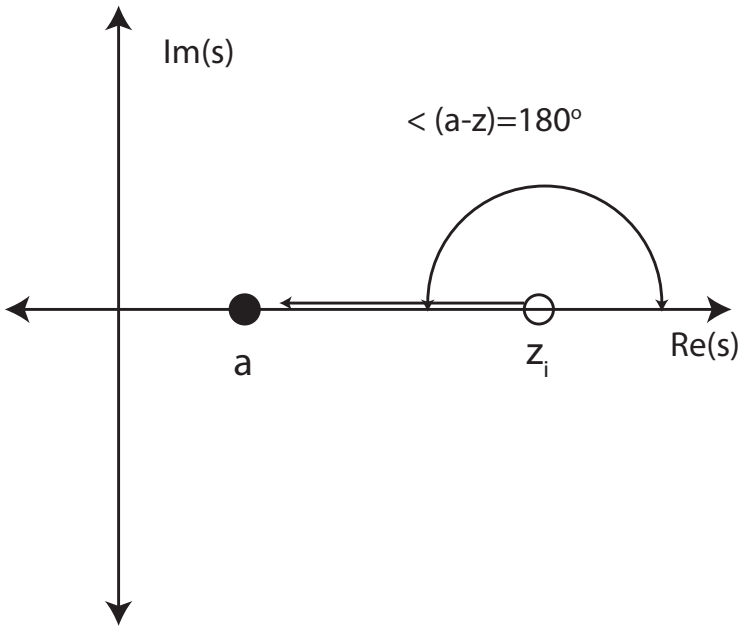
$$\sum_{i=1}^{m} \angle(a - z_i) - \sum_{i=1}^{n} \angle(a - p_i) = 180^\circ \pm n \cdot 360^\circ$$

Phase Contribution from Zeros (z_i):

Case 1: If z_i is on the Real Axis

- If $a > z_i$, then $\angle(a - z_i) = 0^\circ$
  - a is right of the zero.
- If $a < z_i$, then $\angle(a - z_i) = -180^\circ$
  - a is left of the zero.

Contribution is 0° or 180°!



What is the phase of a point on the Real axis? Is it $180°$?

Phase Contribution from Real Poles is also $\pm 180°$

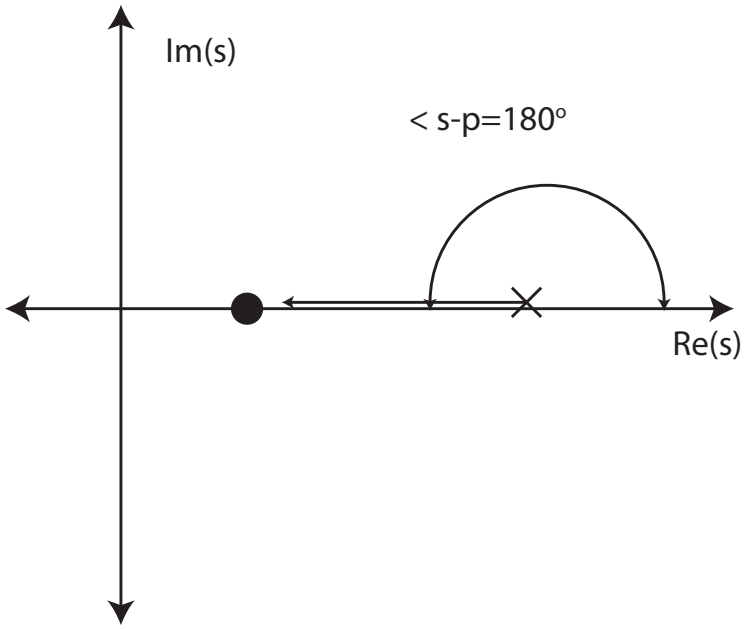
Phase Contribution of Poles:

$$\angle G(a) = \sum_{i=1}^{m} \angle(a - z_i) - \sum_{i=1}^{n} \angle(a - p_i) = -180°$$

Similar to zeros

Case 1: If p_i is Real

- If a is right of p_i, then $\angle(a - p_i) = 0°$
- If a is left of p_i, then $\angle(a - p_i) = -180°$



What is the phase of a point on the Real axis? Is it $180°$?

There is no Phase Contribution from Complex Zeros and Poles!

Case 2: If the z_i is Complex: Complex Roots some in pairs.

$$z_{1,2} = b \pm ci \qquad \sum_{i=1}^{2} \angle(a - z_i) = \angle(a - z_1) + \angle(a - z_2)$$

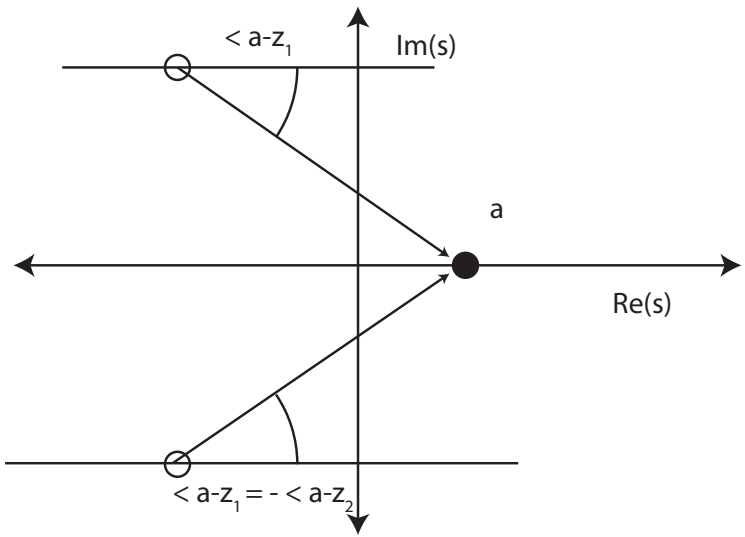
Recall: Angles are defined positive in the CCW direction!

Recall Symmetry: if a is on real axis

$$\angle(a - z_1) = -\angle(a - z_2)$$

There is no contribution to phase on the real axis from a complex pair of zeros!

$$\angle(a - z_1) + \angle(a - z_2) = 0°$$



Likewise, for complex poles, $p_{1,2} = b \pm ci$

$$\angle G(a) = -\sum_{i=1}^{2} \angle(a - p_i) = -\angle(a - p_1) - \angle(a - p_2) = 0°$$

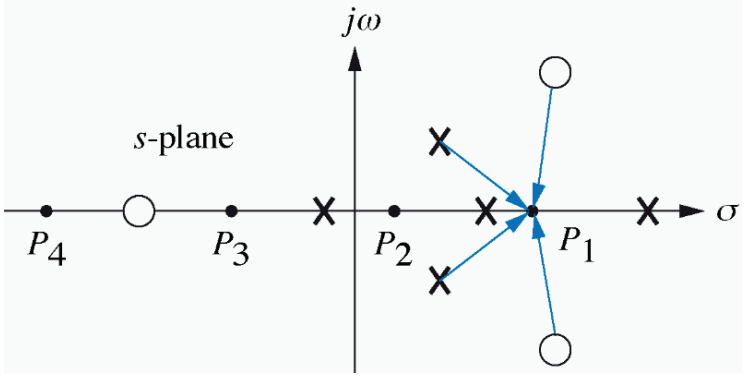
Conclusion: We completely ignore complex poles and zeros!

Root Locus on the Real Axis: A simple rule!

A point on the real axis is on the root locus if it is left of an odd number of poles and zeros

Summary: A point on the real axis: $s = a$

- Complex poles and zeros don't matter
- Real poles and zeros contribute 0° or 180°
  - 0° if the pole/zero is to the left of a
  - 180° if the pole/zero is to the right of a



The **PHASE** of $G(a)$ is

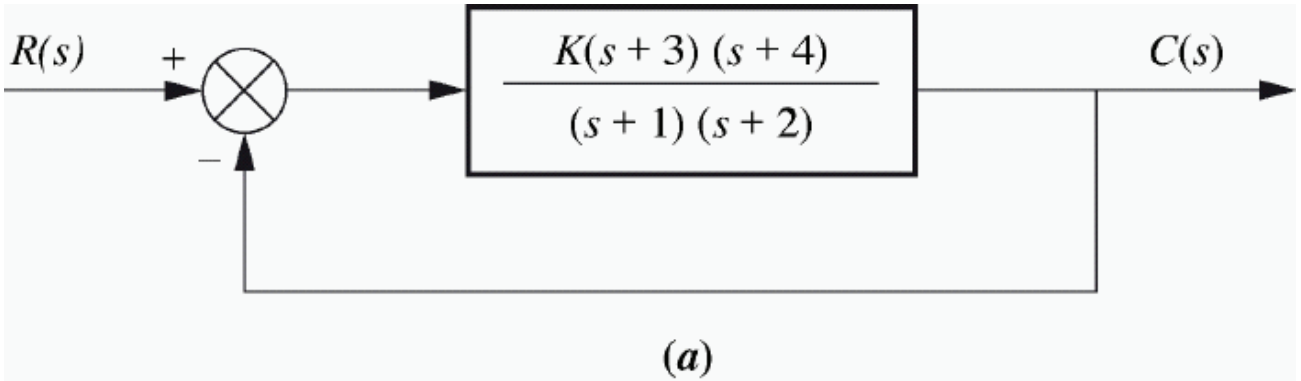
$$\angle G(a) = 180^\circ \cdot (\#\text{ of poles and zeros to the right of } a)$$

A Simple Rule:

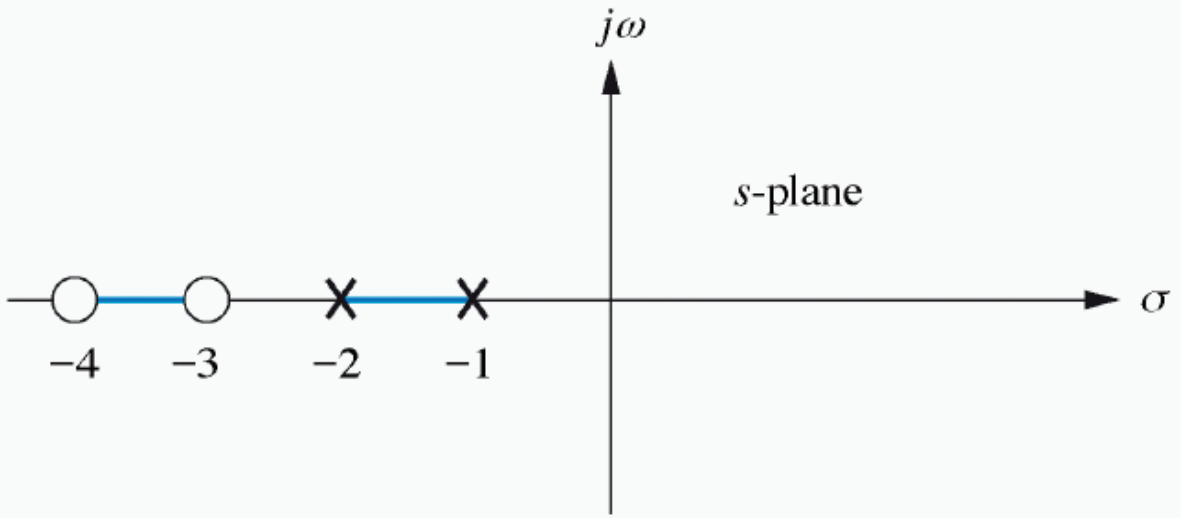
- If the # of poles and zeros to the right of a is EVEN.
  - We are **OFF** the root locus.
- If the # of poles and zeros to the right of a is ODD.
  - We are **ON** the root locus.

Sketching Root Locus on the Real Axis

Example 1



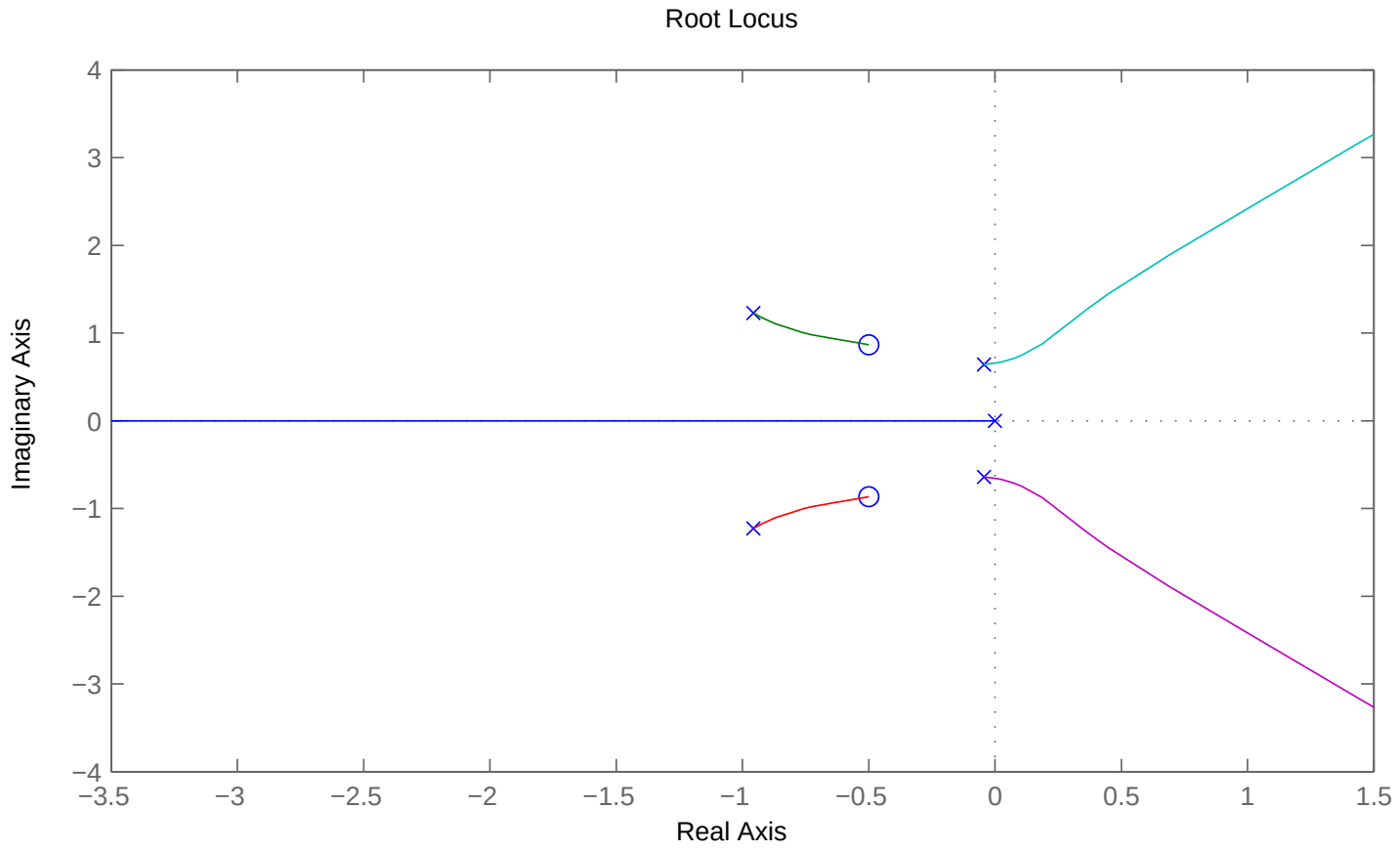
(a)



(b)

Sketching Root Locus on the Real Axis

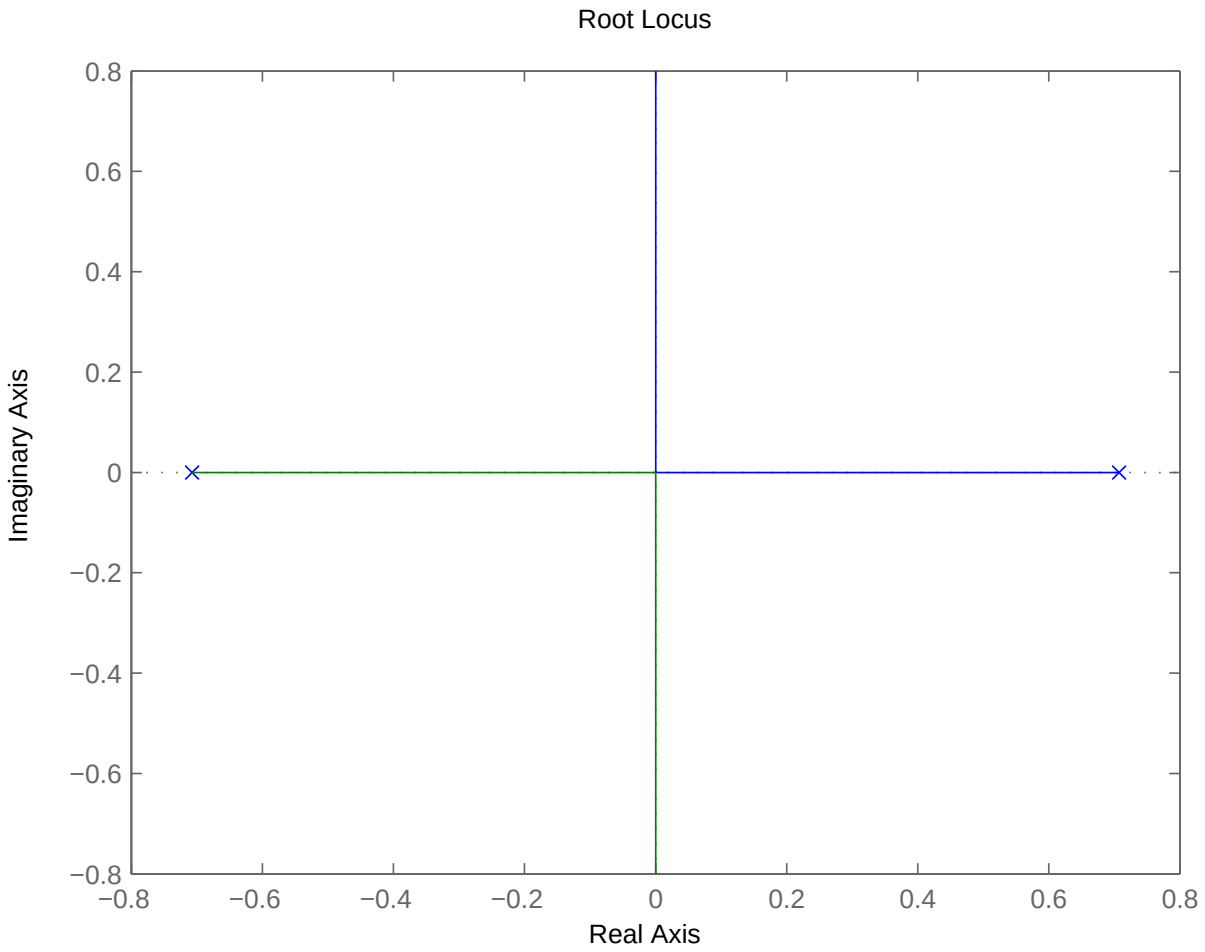
Example 2



Sketching Root Locus on the Real Axis

Example 3

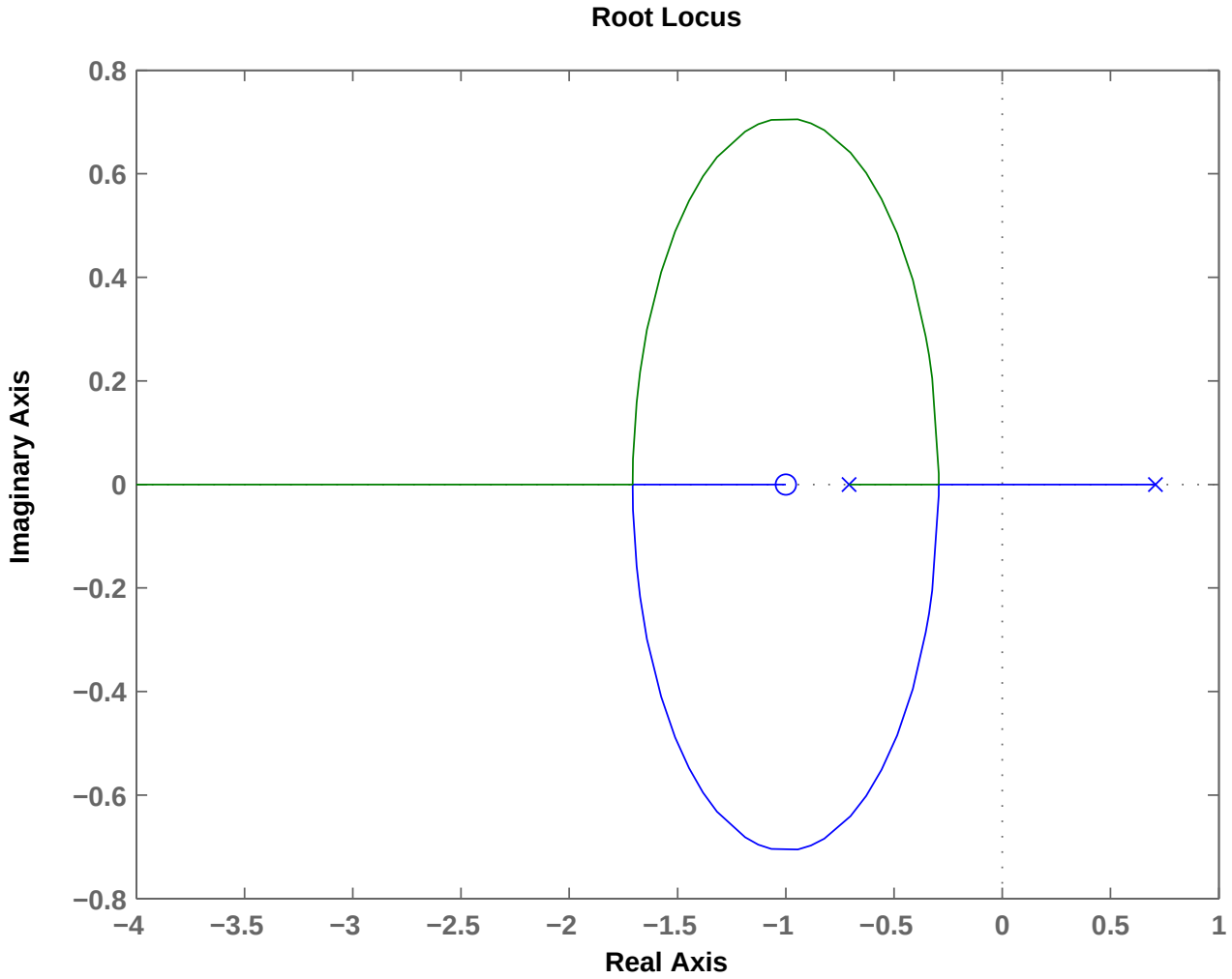
$$G(s) = \frac{1}{s^2 - \frac{1}{2}}$$



Sketching Root Locus on the Real Axis

Example 4

The inverted pendulum with some derivative feedback: $\hat{K}(s) = k(1+s)$

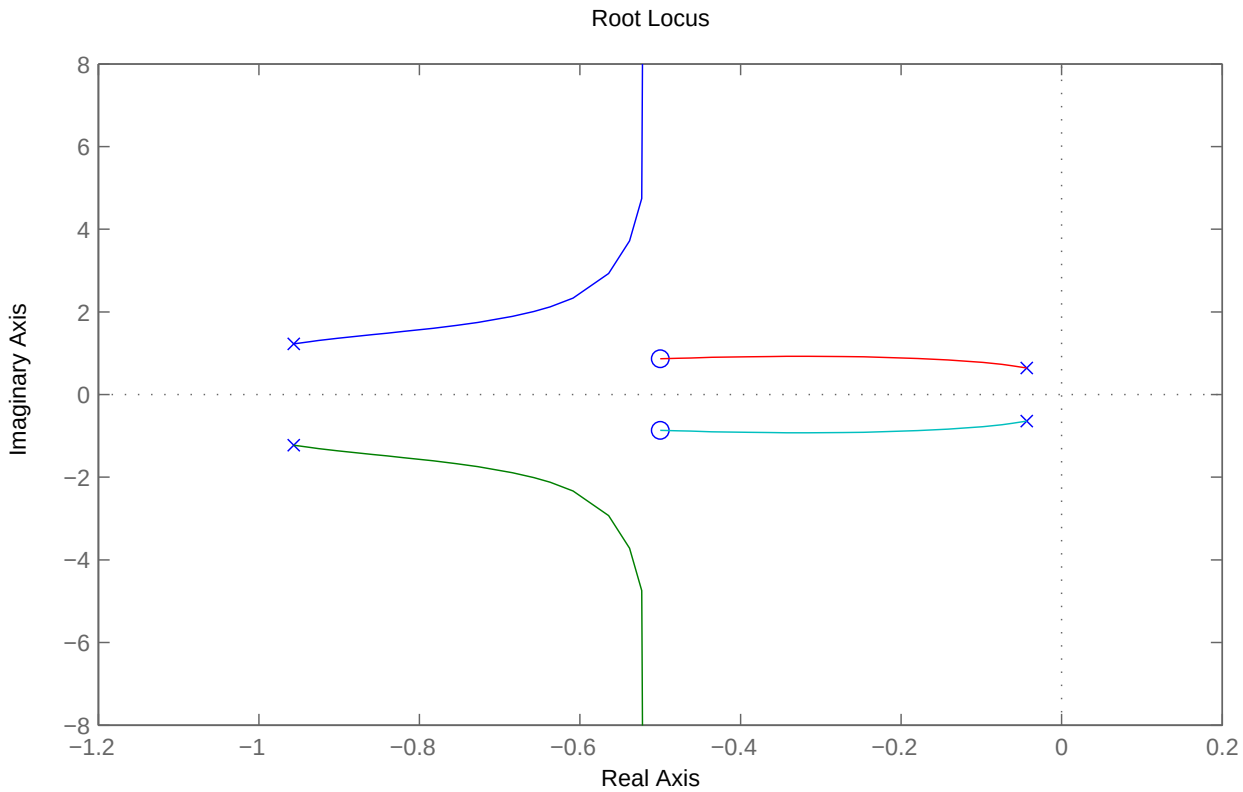


The Root Locus at High Gain

What happens as $k \to \infty$

Now lets look at what happens when gain increases.

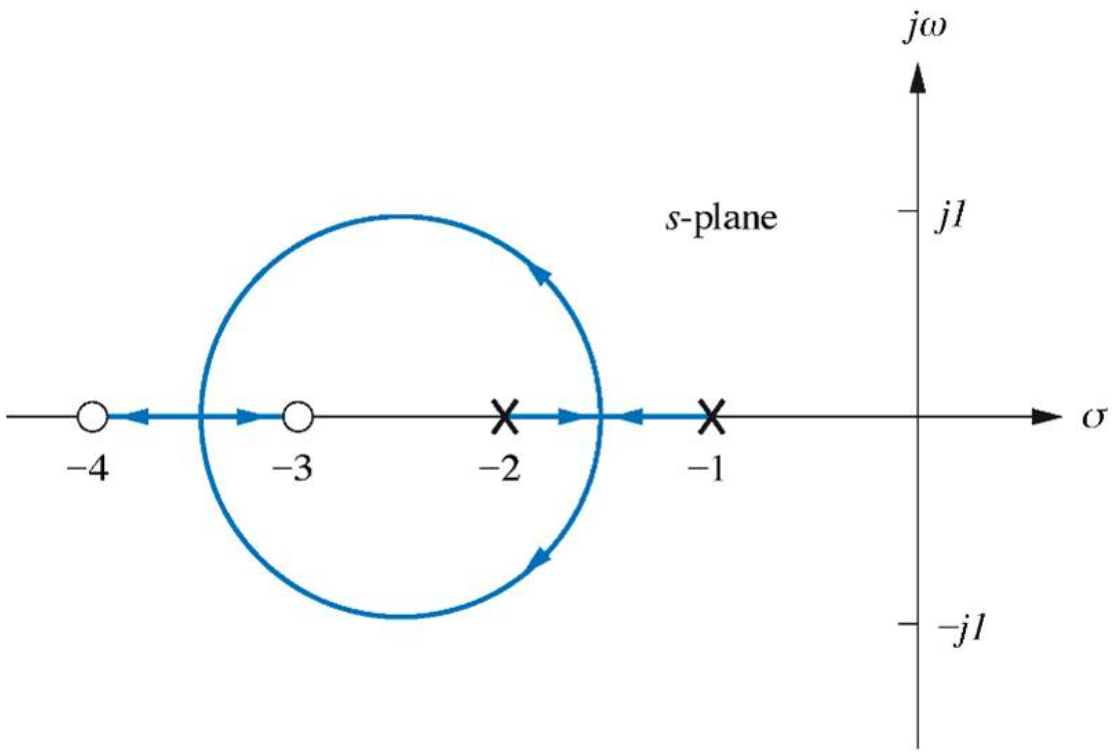
- Recall Continuity and Direction.
- Poles start at open-loop poles (k=0)
- Where do they end up?



Conclusion: Sometimes Poles can behave badly at high gain! (high overshoot)

The Root Locus at High Gain

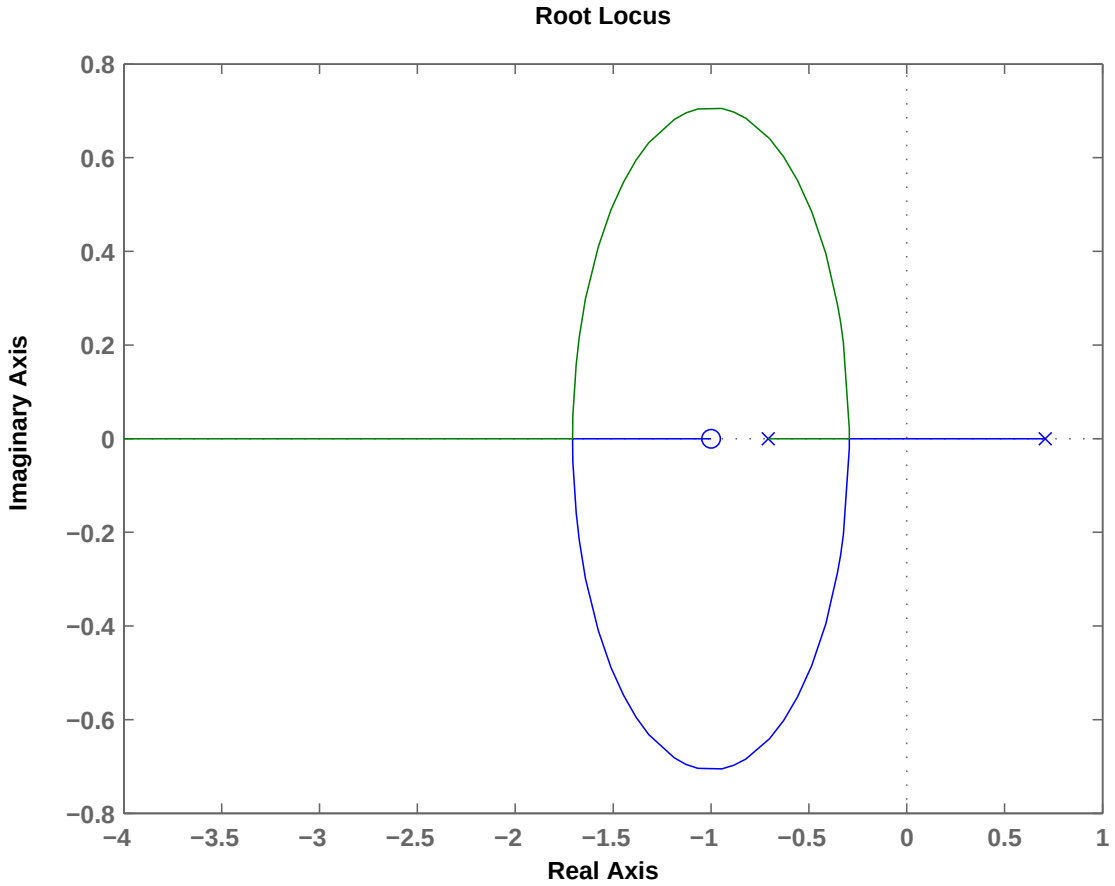
$$G(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)}$$



Conclusion: Sometimes Nothing Much Happens at high gain.

The Root Locus at High Gain

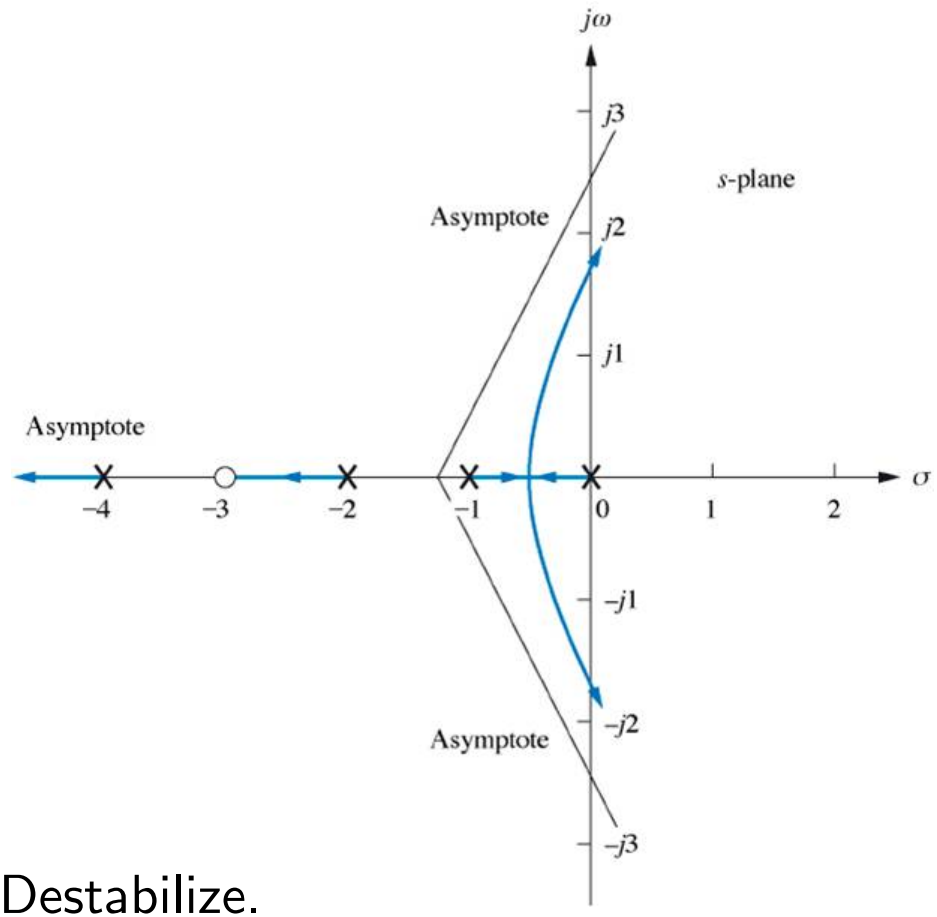
Again, Inverted Pendulum with derivative feedback.



Conclusion: Poles get **More Stable**.

The Root Locus at High Gain

$$G(s) = \frac{s+3}{s(s+1)(s+2)(s+4)}$$



Conclusion: Poles Destabilize.

Notice the **Asymptotes**.

The Root Locus at High Gain

Zeros are on the root locus

So what happens when k is large?

Logically, there are **TWO CASES:**

- Poles can remain small.
- Poles can get big.

Consider the **Small Poles** ($\|s\| < \infty$).

$$G(s) = \frac{n(s)}{d(s)}$$

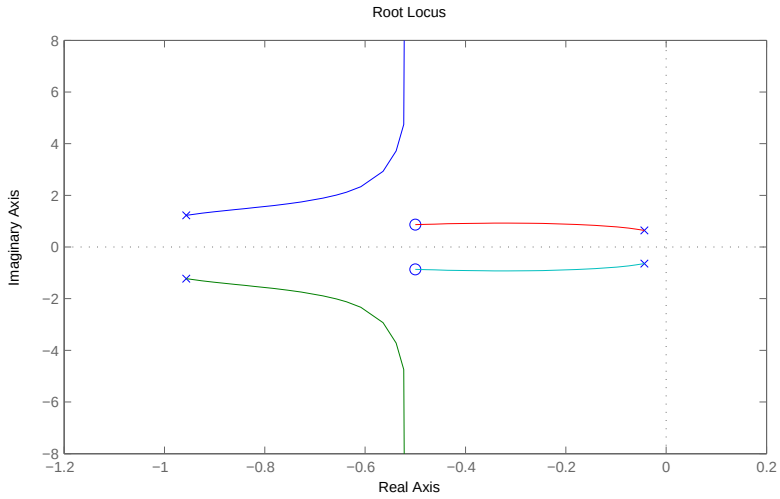
- OL zeros are roots of $n(s) = 0$.
- OL poles are roots of $d(s) = 0$.

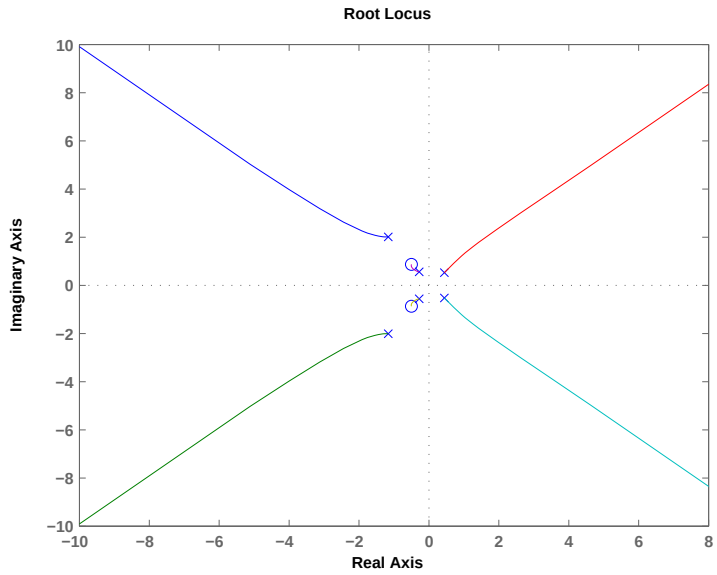
Recall the closed loop:

$$\frac{kG(s)}{1 + kG(s)} = \frac{n(s)}{\frac{1}{k}d(s) + n(s)}$$

CL poles are roots of

$$\frac{1}{k}d(s) + n(s) = 0$$





The Root Locus at High Gain

At high gain, every open-loop Zero attracts a pole

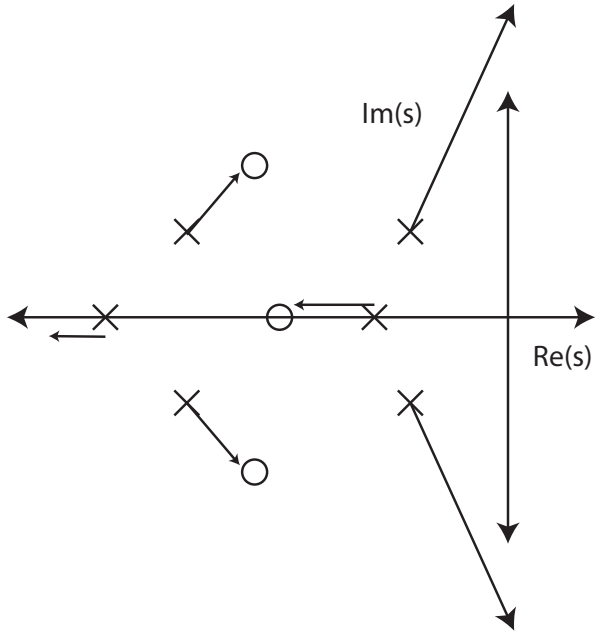
At high gain, small CL poles are roots of

$$\frac{1}{k}d(s) + kn(s) = 0$$

- If s is small, then $d(s)$ is small
- Hence as $k \to \infty$,

$$\frac{1}{k}d(s) + n(s) \cong n(s)$$

- Continuity implies OL zeros are on the root locus.



At high gain, every open-loop Zero attracts a pole.

Problem: What if there are more OL poles than OL zeros?

The Root Locus at High Gain

What happens to the other poles: Asymptotics

Now consider the other possibility:

s also gets Very Large

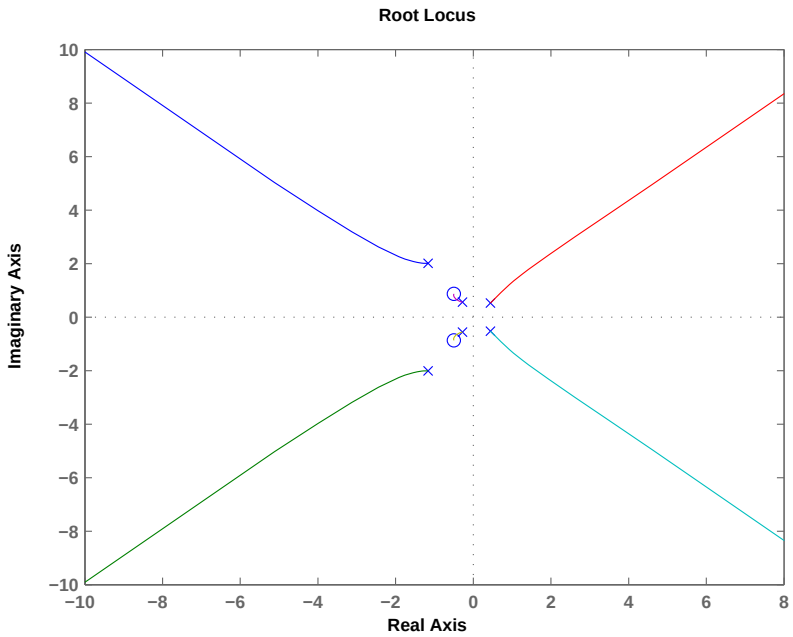
$$d(s) + kn(s) = 0$$

In this case, $d(s)$ is not small.

Very Large solutions of

$$1 + kG(s)$$

are called asymptotics.



- asymptotics increase forever with k
  - $\lim_{k\to\infty} \|s\| = \infty$

Questions:

- Do asymptotes exist?
- Where do they go?

The Root Locus at High Gain

Asymptotics

Consider a point s which is **Very Large**

Recall that a point is on the root locus if

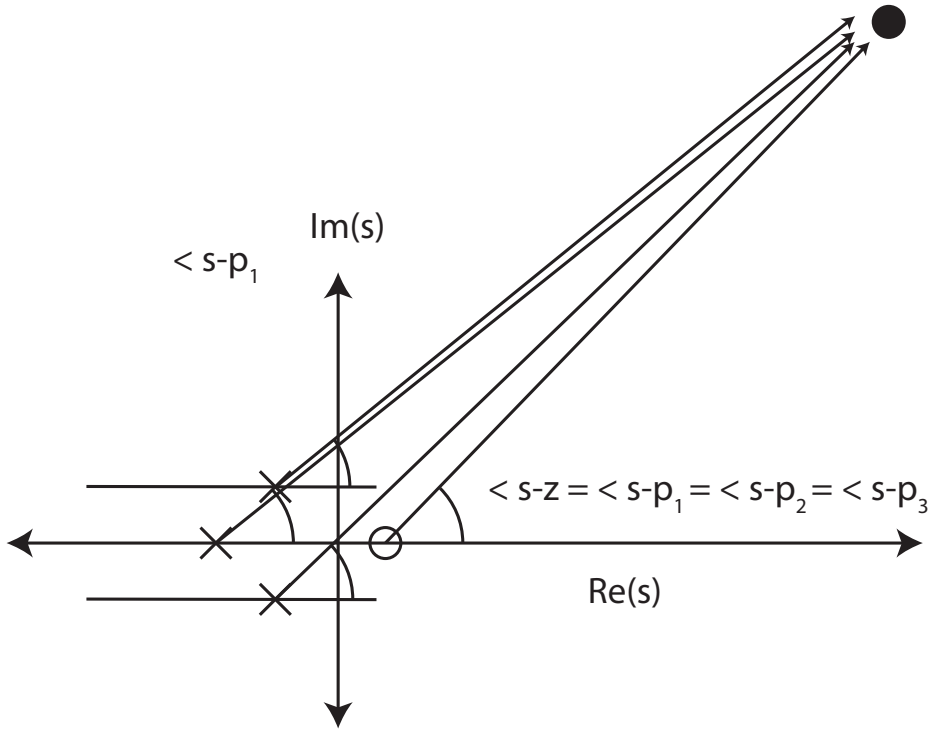
$$\angle G(s) = 180^\circ \pm 360^\circ l$$

Which means:

$$\sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i) = 180^\circ \pm 360^\circ l$$

However, when $\|s\| \to \infty$,

All Angles are the Same!!!



$$\angle(s - z_1) = \angle(s - z_2) = \ldots = \angle(s - z_m)$$
$$= \angle(s - p_1) = \angle(s - p_2) = \ldots = \angle(s - p_n) = \angle_\infty$$

Drawing the Asymptotes

The Number of Asymptotes Is Number of Poles Minus Number of Zeros

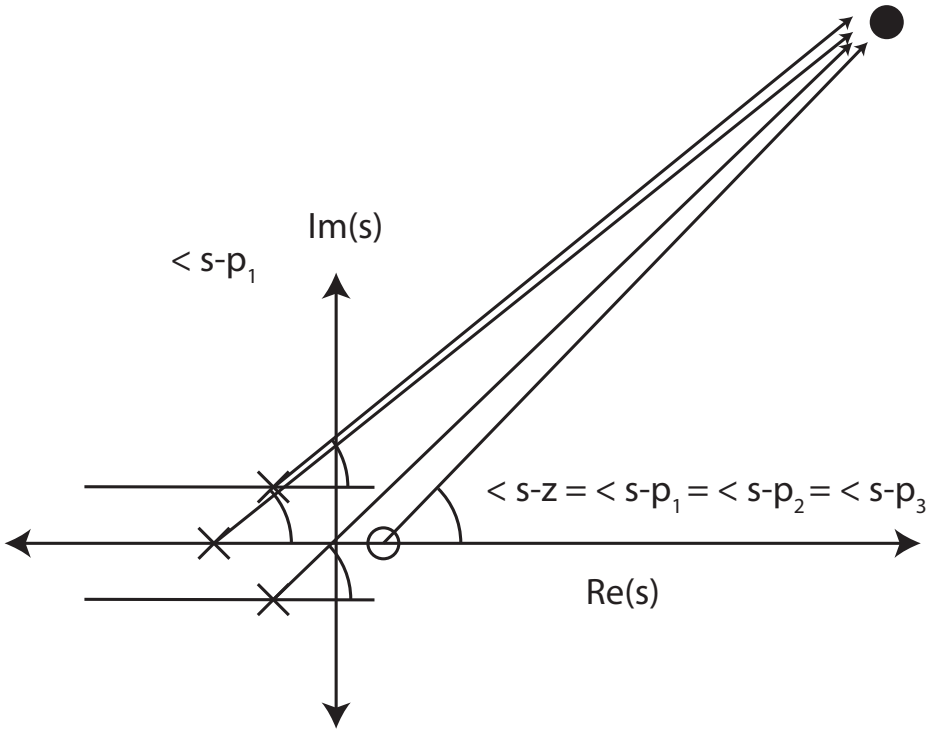
This makes life easier.

- just solve for one angle, $\angle_\infty$.

  $$\angle_\infty \cdot (m - n) = 180^\circ \pm 360^\circ l$$

where

- m is the number of OL zeros
- n is the number of OL poles



So asymptotics occur at

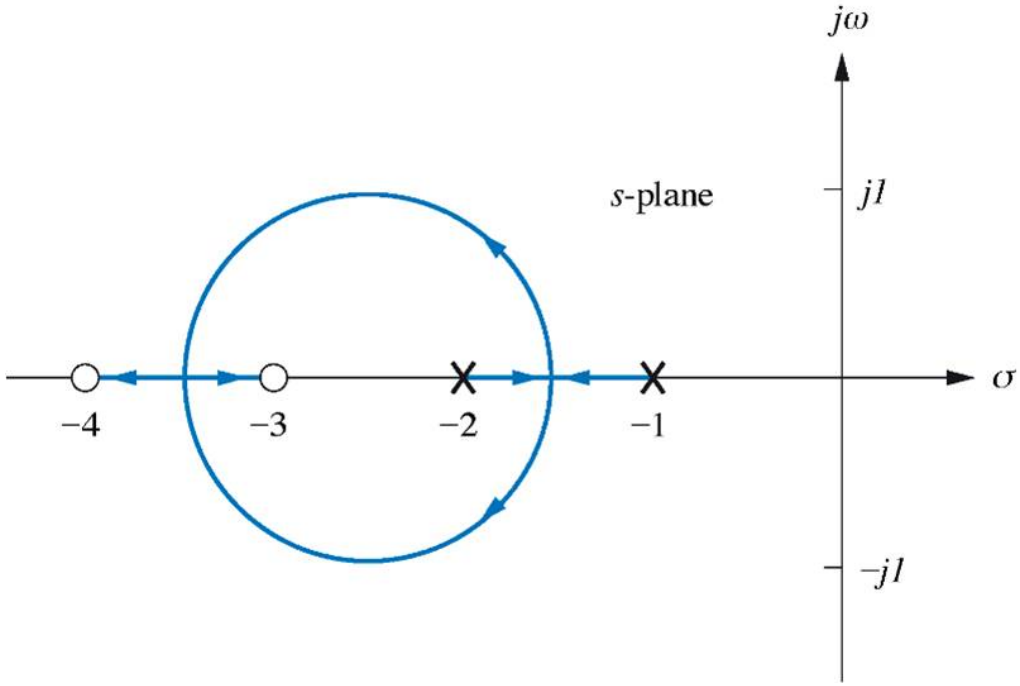
$$\angle_\infty = \frac{1}{m-n}(180^\circ + 360^\circ l)$$

for integers $l = 0, 1, 2, \cdots$.

Asymptotes for Two Poles and Two Zeros

Case 1: $n - m = 0$

$$G(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)}$$



Count: 2 zeros, 2 poles.

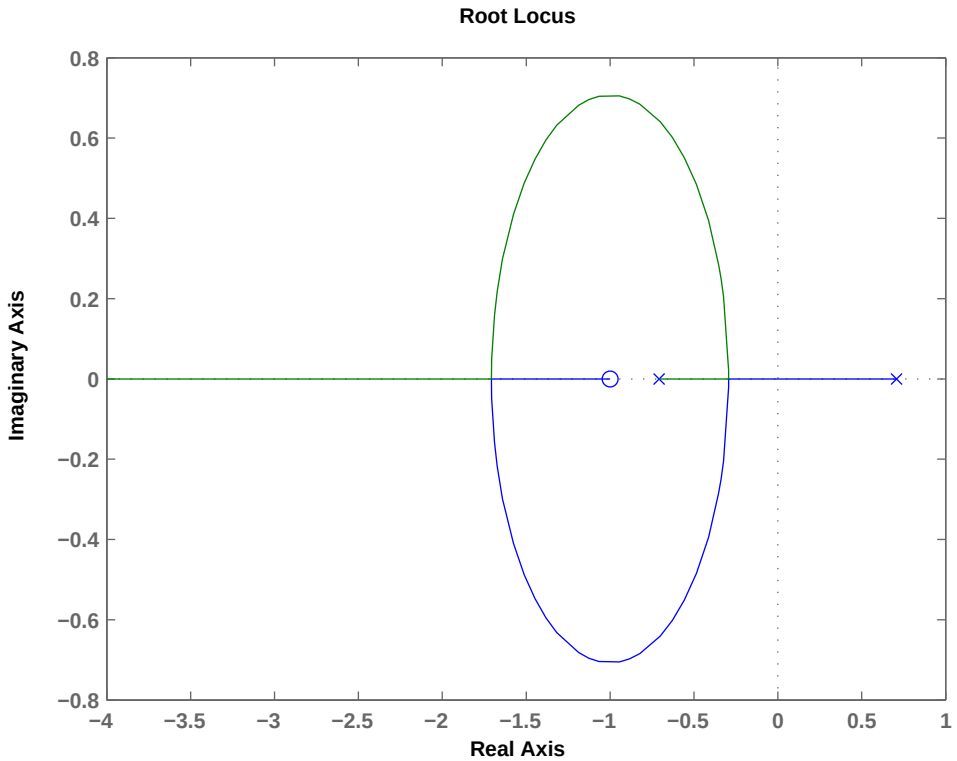
$$m - n = 0 \qquad \angle_\infty = \frac{1}{0}180^\circ = \infty$$

No Asymptotes.

Asymptotes for Two Poles and One Zero

Case 2: $n - m = 1$

$$G(s) = \frac{1}{s^2 - \frac{1}{2}} \qquad K(s) = k(1+s)$$



Count: 1 zeros, 2 poles.

$$m - n = -1 \qquad \angle_\infty = -180^\circ$$

1 asymptote at -180°.

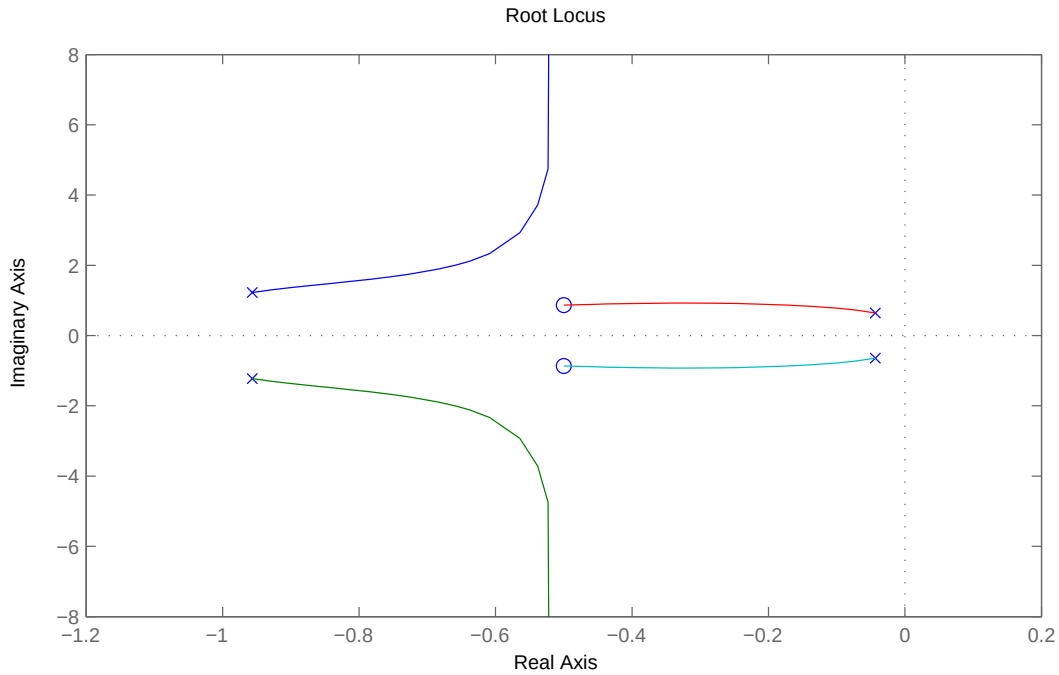
Asymptotes for 4 Poles and 2 Zeros

Case 3: $n - m = 2$

The suspension system.

Count: 2 zeros, 4 poles.

$$m - n = -2$$



$$\angle_\infty = -\frac{1}{2}\left(180^\circ + 360^\circ l\right) = -90^\circ - 180^\circ l = -90^\circ, -270^\circ$$

2 vertical asymptotes at 90° and 270°.

Poles **MAY** destabilize at large gain.

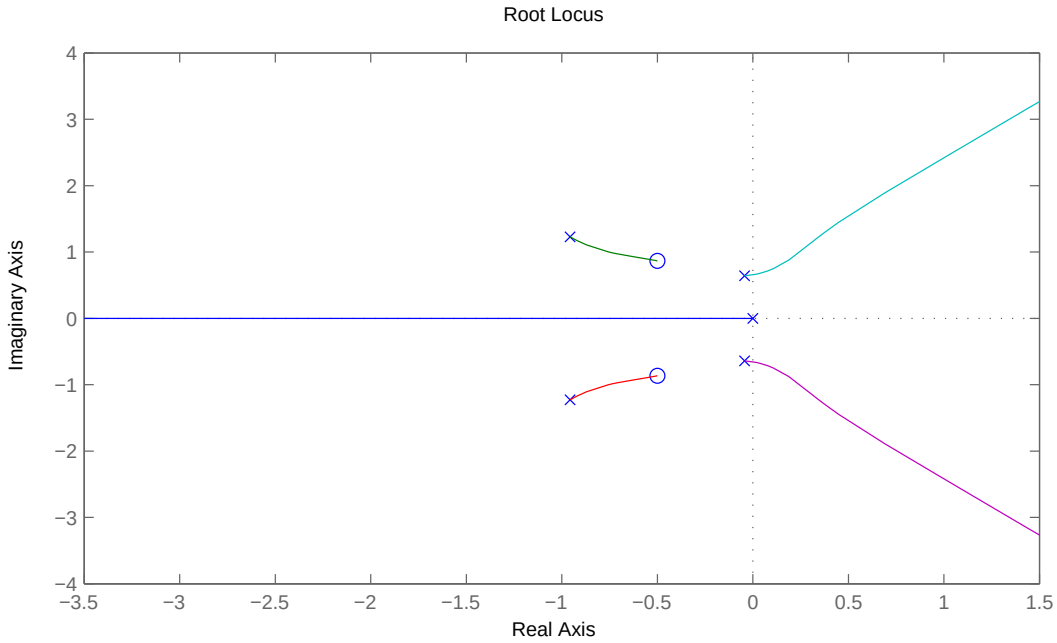
Asymptotes for 5 Poles and 2 Zeros

Case 4: $n - m = 3$

The suspension system with integral Feedback.

Count: 2 zeros, 5 poles.

$$m - n = -3$$



$$\angle_\infty = -\frac{1}{3}\left(180^\circ + 360^\circ l\right) = -60^\circ - 120^\circ l = -60^\circ, -180^\circ, -300^\circ$$

3 asymptotes at 60°, 180° and 300°.

Poles **WILL** destabilize at large gain.

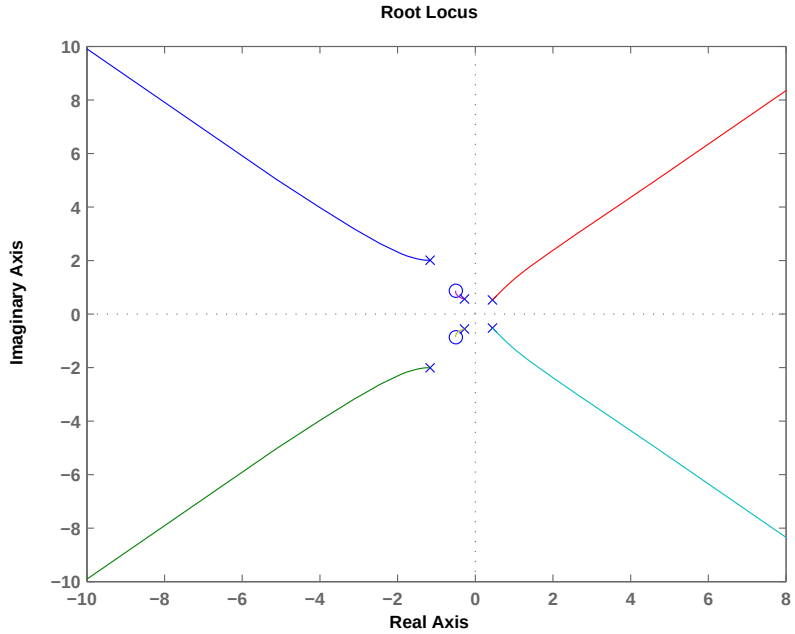
Asymptotes for 6 Poles and 2 Zeros

Case 5: $n - m = 4$

$n(s)$ is degree 2, $d(s)$ is degree 6.

$$G(s) = \frac{s^2 + s + 1}{s^6 + 2s^5 + 5s^4 - s^3 + 2s^2 + 1}$$

Count: 2 zeros, 6 poles.



$$m - n = -4$$

$$\angle_\infty = -\frac{1}{4}(180^\circ + 360^\circ l) = -45^\circ - 90^\circ l = -45^\circ, -135^\circ, -225^\circ, -315^\circ$$

4 asymptotes at 45°, 135°, 225° and 315°.

Poles **WILL** destabilize at large gain.

Asymptotics

Summary

Asymptotes depend only on relative number of poles and zeros.

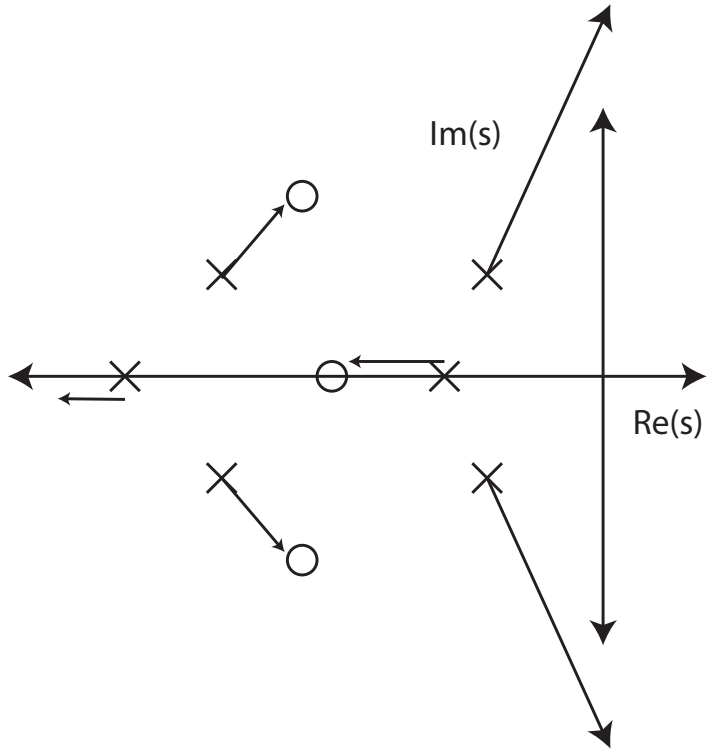
- Location of poles/zeros doesn't matter
  - At least not for the angle

One pole goes to each zero.

When there are more poles than zeros:

Cases:

- $n - m = 0$ - No Asymptotes
- $n - m = 1$ - Asymptote at $180°$
- $n - m = 2$ - Asymptotes at $\pm 90°$
- $n - m = 3$ - Asymptotes at $180°$, $\pm 60°$
- $n - m = 4$ - Asymptotes at $\pm 45°$ and $\pm 135°$



Summary

What have we learned today?

Review

- Definition of Root Locus

Points on the Real Axis

- Symmetry
- Drawing the Real Axis

What Happens at High Gain?

- The effect of Zeros
- Asymptotes

Next Lecture: Centers of Asymptotes, Break Points and Departure Angles